AP PRECALCULUS

Unit 1 • Lesson 1.12 practice • Answer key

Lesson 1.12 • Transformations of Functions

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How is the graph of $g(x) = f(x+3) - 1$ obtained from the graph of f?

(A) Right 3, down 1
(B) Left 3, down 1
(C) Left 3, up 1
(D) Right 3, up 1

Solution

Answer: (B). $x + 3 = x - (-3)$: a shift of 3 units left; -1: down 1.

Question 2

NO CALCULATOR

How is the graph of $g(x) = f(2x)$ obtained from the graph of f?

(A) Vertical dilation by a factor of 2
(B) Horizontal dilation by a factor of 2
(C) Horizontal dilation by a factor of $\frac{1}{2}$
(D) Vertical dilation by a factor of $\frac{1}{2}$

Solution

Answer: (C). $f(bx)$ is a horizontal dilation by a factor of $\frac{1}{b} = \frac{1}{2}$.

Question 3

NO CALCULATOR

The point $(4, -2)$ is on the graph of f. Which point must be on the graph of $y = 3f(x-1) + 2$?

(A) $(3, -4)$
(B) $(5, -6)$
(C) $(5, -4)$
(D) $(4, -4)$

Solution

Answer: (C). x moves right 1 to 5; y becomes $3(-2) + 2 = -4$.

Question 4

NO CALCULATOR

How is the graph of $y = -f(x)$ obtained from the graph of f?

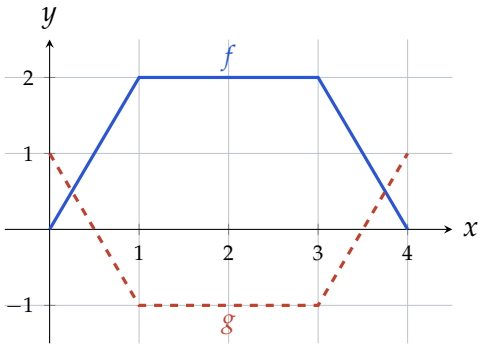
(A) Reflection over the x-axis
(B) Reflection over the y-axis
(C) Reflection over the line $y = x$
(D) Shift down 1 unit

Solution
Answer: (A). Each output is multiplied by -1.

Question 5 (graph)

NO CALCULATOR

The graphs of f (solid) and g (dashed) are shown. Which is an equation for g in terms of f?



(A) $g(x) = -f(x) - 1$
(B) $g(x) = f(-x) + 1$
(C) $g(x) = -f(x) + 1$
(D) $g(x) = -f(x + 1)$

Solution
Answer: (C). Reflect f over the x-axis and shift up 1: $(0,0) \to (0,1)$, $(1,2) \to (1,-1)$, $(3,2) \to (3,-1)$, $(4,0) \to (4,1)$.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = x^2$ and $g(x) = -2(x-1)^2 + 8$.

(a) Describe the transformations that take the graph of f to the graph of g.
(b) Find the zeros of g.
(c) If f is restricted to the domain $[-2, 2]$ (range $[0, 4]$), find the corresponding domain and range of g.

Solution
(a) Shift right 1 unit, vertical dilation by a factor of 2, reflection over the x-axis, then shift up 8 units.
(b) $-2(x-1)^2 + 8 = 0 \iff (x-1)^2 = 4 \iff x = 3$ or $x = -1$.
(c) Domain: $[-2+1,\ 2+1] = [-1, 3]$. Range: $-2 \cdot [0, 4] = [-8, 0]$, then $+8$: $[0, 8]$.